"

On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."



Q3

v is a unit vector in R^n

H is a $n \times n$ matrix

$H = I - 2vv^T$

As v is a unit vector

$\|v\| = 1$

a)

$H^2 = H.H = (I - 2vv^T)(I - 2vv^T)$

$= I^2 - I.2vv^T - 2vv^T.I + 4vv^T.vv^T$

$= I(I - 2vv^T - 2vv^T) + 4v.\|v\|^2.v^T$

$= I - 2vv^T - 2vv^T + 4v.\|v\|^2.v^T$

$\|v\|^2 = v^T.v$

$\|v\| = \pm\sqrt{v^T.v}$ $\Rightarrow \|v\| = 1$

So, $H^2 = I - 4vv^T + 4vv^T$

$H^2 = I - 4(1) + 4(1)$

$\boxed{H^2 = I}$

b.) $HH^T = (I - 2vv^T)(I - 2vv^T)^T$

$= (I - 2vv^T)(I^T - (2vv^T)^T)$

$= (I - 2vv^T).(I - 2(v^T)^T v^T)$

$= (I - 2vv^T).(I - 2vv^T)$

$= I^2 - 2vv^T - 2vv^T + 4vv^T.vv^T$

$= I - 4vv^T + 4v\|v\|^2 v^T$

$= I - 4vv^T + 4vv^T$

$= I$

$H.H^T = I$

$H.H^T = H^T.H = I \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

H is orthogonal matrix

c.) For a matrix A, a vector (non zero) will be eigen-vector of A corresponding eigen value λ if, $\boxed{AV = \lambda V}$

now, $Hv = (I - 2vv^T).v$

$= Iv - 2vv^T v$

$= v - 2v\|v\|^2$

As I is identity matrix, so 1 is the only eigen value of I and $Iv = 1.v = v$ and $\|v\| = +\sqrt{v^T.v}$

v is a unit vector, so, $\|v\| = 1 = +\sqrt{v^T v}$ and hence, $v^T v = \|v\|^2 = 1$

$Hv = v - 2v.1 = -v$

$\Rightarrow Hv = -v = (-1).v$

So, we have $\Rightarrow Hv = \lambda v$ where $\lambda = -1$

So, v is an eigen vector of H and corresponding eigen value for v is $= \lambda = -1$

in the beginning tank A $10(0) = 10L$

tank B $w(0) = 40L$

$$\begin{bmatrix} 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} c_2$$

t sıfır ran

$10 = c_1 + c_2$

$40 = c_1 - c_2$

$50 = c_1$

$\boxed{c_1 = 25}$

$c_2 = -15$

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0t} \cdot 25 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} \cdot (-15)$$

$$v(1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 25 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2} (-15)$$

$25 + (-15) \cdot e^{-2}$ => for A => %45

$25 + 15e^{-2}$ = for B => %55

$$v+1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 25 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cancel{e^{-26}}^{0} (-15)$$

%25 liquid left in tank A

%15 liquid left in tank B

Q2

A 1 $x=y$ 3-B ibir cizgidir ve 3 boyutlu bir cisimle kesiscebilir. Bu yuzden 4 3-D. Daha fazla boyut icerisinde de hacimden bahsedebiliriz ama minimum 3-D icerisinde olmak kaydiyla

b) eger iki plane birlesirse, buna dikdortgen prizması diyebiliriz.